Package 'PACEanalysis'

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Type Package

Title Analysis of PACE DNA Methylation Microarray Data

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Author Alexandra Binder

Maintainer Alexandra Binder <ambinder@hawaii.edu>

Description Reads in IDAT files, performs data quality checks and performs site-specific data analysis for a specified exposure or outcome of interest

License What license is it under?

Encoding UTF-8

LazyData true

RoxygenNote 7.2.0

Suggests knitr, rmarkdown

VignetteBuilder knitr

Imports methods,
reshape,
lmtest,
GGally,
sva,
gplots,
growthstandards,
heatmap.plus,
pvclust,
EpiDISH,
sesame,
sesameData,
wateRmelon,
RPMM,
RefFreeEWAS,
SummarizedExperiment,
IlluminaHumanMethylationEPICmanifest,
IlluminaHumanMethylation450kmanifest,
FlowSorted.Blood.EPIC,
FlowSorted.CordBloodCombined.450k,
ewastools,
matrixStats,
dplyr

Depends R (>= 4.1),
minfi,
ggplot2,
Hmisc,
MASS,
sandwich,
lmtest,
parallel,
doParallel,
plyr,
IlluminaHumanMethylation450kanno.ilmn12.hg19,
IlluminaHumanMethylationEPICanno.ilm10b4.hg19

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`dataAnalysis` *Analyze the Data*

Description

The goal is to output the information necessary for meta-analysis

Usage

```
dataAnalysis(
  phenofinal = NULL,
  betafinal = NULL,
  array = "EPIC",
  Table1vars = c("Age", "BMI", "Smoke", "Parity", "Sex", "Gestage", "BWT"),
  StratifyTable1 = FALSE,
  StratifyTable1var = NULL,
  maxit = 100,
  vartype = "ExposureCont",
  varofinterest = NULL,
  adjustmentvariables = c("Age", "Gestage"),
  Omega = NULL,
  RestrictToSubset = FALSE,
  RestrictionVar = NULL,
  RestrictToIndicator = NULL,
  robust = TRUE,
  RunUnadjusted = TRUE,
  RunAdjusted = TRUE,
```

```
  RunCellTypeAdjusted = TRUE,
  RunSexSpecific = TRUE,
  RunCellTypeInteract = TRUE,
  runparallel = FALSE,
  number_cores = NULL,
  destinationfolder = NULL,
  savelog = TRUE,
  cohort = NULL,
  analysisdate = NULL,
  analysisname = NULL
)
```

Arguments

phenofinal	A data.frame of phenotype data that is subset to the samples among the matrix of processed beta-values. Be sure that the dataframe phenofinal includes the column 'Sex', which can only take three possible character values: 'Female', 'Male', or 'Unknown'. Missing values are not allowed. The data.frame must also have the column 'Basename'. If RestrictToSubset is TRUE, this data.frame must also include the column specified in the argument RestrictionVar
betafinal	Matrix of the processed beta-values
array	A character string of array type; options: "EPIC" or "450K"
Table1vars	An character vector of variable column names to include in the descriptive Table 1. The column names must correspond the column names in the pData DataFrame of the GenomicRatioSet (e.g. c("Age", "BMI", "Smoke")); see the function loadingSamples for the new column names
StratifyTable1	Logic statement (TRUE or FALSE), whether to stratify the Table 1 data on a specific variable like infant sex or case/control status; default is FALSE
StratifyTable1var	A character string indicating the column name for the variable on which to stratify the Table 1 data; default is FALSE. Must be specified if StratifyTable1 is TRUE
maxit	If running robust linear regression models, the maximum number of iterations (default=100)
vartype	A character string indicating how to model DNA methylation; four options: "ExposureCont", "ExposureCat", "OutcomeCont", "OutcomeBin". If "ExposureCont", the variable of interest (varofinterest) is modeled as a continuous exposure and assumed to be numeric. If "ExposureCat", the variable of interest is modeled as a categorical exposure and assumed to be a factor. If "OutcomeCont", the variable of interest is modeled as the outcome, and assumed to be continuous. If "OutcomeBin", the variable of interest is modeled as the outcome, and assumed to be binary (values are 0 or 1). If DNAm is modeled as the exposure, the site-specific effect estimates will be for a 0.01 increase in beta-value
varofinterest	A character string of the column name of the exposure or outcome of interest (e.g. "BWT"). The column name must correspond to a column name in the pData DataFrame of the GenomicRatioSet; see the function loadingSamples for the new column names
adjustmentvariables	A character vector of variable column names to include in your multivariable adjusted models (e.g. c("Age","Gestage")). The column names must correspond

the column names in the pData DataFrame of the GenomicRatioSet; see the function loadingSamples for the new column names

`Omega` A required matrix of cell composition estimates generated by the preprocessingofData function

`RestrictToSubset`
Logic statement (TRUE or FALSE), whether to restrict the analysis to a specific subset of the samples. The default is FALSE, in other words, do not restrict the analysis to a specific subset of the samples. If specified TRUE, must provide a character string to the arguments RestrictionVar and RestrictToIndicator; see those arguments for additional details.

`RestrictionVar` A character string indicating the column name for the variable in the phenofinal data.frame that will be used to subset the data. Must be specified if RestrictToSubset is TRUE; default is NULL

`RestrictToIndicator`
A character string indicating the category of the RestrictionVar you would like to restrict the analysis to. This character string must equal one of the RestrictionVar categories. This argument is only required if RestrictToSubset = TRUE; default is NULL

`robust` Logic statement (TRUE or FALSE), whether to run robust linear regressions. The default is TRUE

`RunUnadjusted` Run unadjusted models? Logic statement (TRUE or FALSE), default = TRUE

`RunAdjusted` Run adjusted models? Logic statement (TRUE or FALSE), default = TRUE

`RunCellTypeAdjusted`
Run adjusted analysis, including cell composition? Logic statement (TRUE or FALSE), default = TRUE

`RunSexSpecific` Run models stratifying by infant sex? Logic statement (TRUE or FALSE), default = TRUE

`RunCellTypeInteract`
Whether to run CellDMC to identify the specific cell type(s) driving the differential methylation. Logic statement (TRUE or FALSE), default = TRUE

`runparallel` Whether to run the site-specific analysis in parallel (TRUE or FALSE), default = FALSE

`number_cores` If running in parallel, a numeric string indicating the number of cores to run (e.g. 4), default = NULL

`destinationfolder`
A character string indicating the location where files should be saved, e.g. "C:\Home\PACE\BirthSize"

`savelog` logic; TRUE indicates to save a log of the functions run (default)

`cohort` A character string for the cohort's acronym (e.g. "HEBC")

`analysisdate` A character string indicating the date the analysis was run. Please specify in the form: YEARMONTHDAY, e.g. "20200205" for February 5th 2020

`analysisname` An optional character string for the analysis name. This will be used to label the output directory and the RData files that contain the final results. This can be helpful to specify if running additional sensitivity analyses

Details

Data analysis steps include:

1. Create a Table 1 descriptive table for the cohort

2. Test the association between estimated cellular components and the exposure variable, the adjustment variables, and batch (if included)
3. Perform association analysis between variable of interest and each CpG site individually. If vartype is "ExposureCont", "ExposureCat", or "OutcomeCont", the association is estimated using linear regression modelling (if robust=TRUE, using the rlm() function in R; otherwise, using lm()). If vartype is "OutcomeBin", a logistic regression is used to estimate the association (glm(family="binomial")). Analyses performed include:
    - Unadjusted analysis if RunUnadjusted=TRUE
    - Analysis adjusting for specified adjustment variables if RunAdjusted=TRUE
    - Analysis adjusting for specified adjustment variables and estimated cell composition if RunCellTypeAdjusted=TRUE
    - If RunSexSpecific=TRUE, RunUnadjusted=TRUE, and there are at least 10 female infants in the dataset, unadjusted analysis among the female infants
    - If RunSexSpecific=TRUE, RunAdjusted=TRUE, and there are at least 10 female infants in the dataset, analysis adjusting for specified adjustment variables among the female infants
    - If RunSexSpecific=TRUE, RunCellTypeAdjusted=TRUE, and there are at least 10 female infants in the dataset, analysis adjusting for specified adjustment variables and estimated cell composition among the female infants
    - If RunSexSpecific=TRUE, RunUnadjusted=TRUE, and there are at least 10 male infants in the dataset, unadjusted analysis among the male infants
    - If RunSexSpecific=TRUE, RunAdjusted=TRUE, and there are at least 10 male infants in the dataset, analysis adjusting for specified adjustment variables among the male infants
    - If RunSexSpecific=TRUE, RunCellTypeAdjusted=TRUE, and there are more than 10 male infants in the dataset, analysis adjusting for specified adjustment variables and estimated cell composition among the male infants
    - If RunCellTypeInteract=TRUE, output of CellDMC (from package EpiDISH) based on the adjusted model

Value

Outputs a list of data frames with the site-specific association results. Also creates a new folder in the Output folder for the specified exposure/outcome variable. The folder includes:

1. Descriptive statistics of cohort (Table 1; csv file)
2. Figures (PDFs/PNGs) of associations, including:
    - QQplot for each analysis
    - Volcano plot for each analysis
    - Scatterplot for the 20 most significant associations for each analysis
3. An RData file that includes a list of data frames with the site-specific association results
4. An RData file that includes a list of the lambdas for each analysis

Examples

```
## Not run:
allresults<-dataAnalysis(phenofinal=pData(processedOut$mset),
                         betafinal=processedOut$processedBetas,
                         array="EPIC",
                         Omega=processedOut$Omega,
                         StratifyTable1=TRUE,
```

```
                        StratifyTable1var="Sex",
                        maxit=100,
                       Table1vars=c("Age","BMI","Smoke","Parity","Sex","Gestage","BWT"),
                        vartype="ExposureCont",
                        varofinterest="BWT",
                        adjustmentvariables=c("Age","Gestage"),
                        destinationfolder="H:/UCLA/PACE/Birthweight-placenta",
                        robust=TRUE,
                        RunUnadjusted=TRUE,
                        RunAdjusted=TRUE,
                        RunCellTypeAdjusted=TRUE,
                        RunSexSpecific=TRUE,
                        RunCellTypeInteract=TRUE,
                        RestrictToSubset=FALSE,
                        RestrictionVar=NULL,
                        RestrictToIndicator=NULL,
                        runparallel=FALSE,
                        number_cores=NULL,
                        savelog=TRUE,
                        cohort="HEBC",analysisdate="20200710",
                        analysisname="primaryBWT")

## End(Not run)
```

`detectionMask` *Detection Masking*

Description

Masks beta-values based on poor intensity values

Usage

```
detectionMask(
  processedBetas = NULL,
  DetectionPvals = NULL,
  DetectionPvalCutoff = 0.05,
  IndicatorGoodIntensity = NULL,
  cohort = NULL,
  analysisdate = NULL,
  destinationfolder = NULL
)
```

Arguments

`processedBetas`	The preprocessed beta-value matrix
`DetectionPvals`	An optional matrix of detection p-values
`DetectionPvalCutoff`	a numeric value indicating the detection p-value cut-off (default=0.05)
`IndicatorGoodIntensity`	An optional matrix used to mask methylation values with a poor intensity value based on the number of beads and/or intensity values of zero; NA if poor intensity value and 1 otherwise. Output from ExploratoryDataAnalysis function

cohort	A character string for the cohort's acronym (e.g. "HEBC")
analysisdate	A character string indicating the date the analysis was run. Please specify in the form: YEARMONTHDAY, e.g. "20200205" for February 5th 2020
destinationfolder	A character string indicating the location where files should be saved, e.g. "C:\Home\PACE\BirthSize"

Details

Masks beta-values based on the detection p-value; the default p-value cut-off is 0.05.

Value

Returns and saves an RData file of the preprocessed beta-value matrix with beta-values masked based on poor intensity values. Also saves a csv of the CpG distributions after this masking

Examples

```
## Not run:
betasabovedetection<-detectionMask(processedBetas=processedOut$processedBetas,
                                   DetectionPvals=EDAresults$DetectionPval,
                                   DetectionPvalCutoff=0.05,
                                 IndicatorGoodIntensity=EDAresults$IndicatorGoodIntensity,
                                   destinationfolder="H:/UCLA/PACE/Birthweight-placenta",
                                   cohort="HEBC",analysisdate="20200710")

## End(Not run)
```

ExploratoryDataAnalysis

Exploratory Data Analysis

Description

Exploratory data analysis to identify possible sample swaps and technical outliers

Usage

```
ExploratoryDataAnalysis(
  RGset = NULL,
  globalvarexplore = NULL,
  DetectionPvalMethod = NULL,
  DetectionPvalCutoff = 0.05,
  minNbeads = 0,
  FilterZeroIntensities = FALSE,
  destinationfolder = NULL,
  savelog = TRUE,
  cohort = NULL,
  analysisdate = NULL
)
```

Arguments

RGset	The RGset generated by the function loadingSamples. The pData data.frame requires a column called "ID", which is a unique identifier for each sample. The pData data.frame also requires a column called "Sex", which can only take three possible character values: 'Female', 'Male', or 'Unknown'.
globalvarexplore	A vector of characteristic column names to explore in association with the first 12 principle components. These column names must be in the pData frame of the RGset specified by the RGset argument
DetectionPvalMethod	A character string indicating the detection p-value method to use; options include: SeSAME method ("SeSAMe"; default) or ewastools method ("ewastools").
DetectionPvalCutoff	a numeric value indicating the detection p-value cut-off (default=0.05)
minNbeads	a numeric value indicating the minimum number of beads to include a beta-value (default=0; a value of 0 corresponds to no filtering)
FilterZeroIntensities	a logic statement (TRUE/FALSE; default=FALSE).
destinationfolder	A character string indicating the location where files should be saved, e.g. "C:\Home\PACE\BirthSize'
savelog	logic; TRUE indicates to save a log of the functions run (default)
cohort	A character string for the cohort's acronym (e.g. "HEBC")
analysisdate	A character string indicating the date the analysis was run. Please specify in the form: YEARMONTHDAY, e.g. "20200205" for February 5th 2020

Details

The sesame package is used to calculate the detection p-values. Samples to exclude based on whether they look like a sex mix-up, and poor detection p-values (detection p-value>0.05) for more than 5 percent of probes.

Value

PDFs/PNGs of exploratory data analysis in "EDA" subfolder of "Output" folder, including: red/green signal intensities, distribution of intensity across all sites, heatmap of chrY, signal intensities on X and Y chromosomes, dendogram and heatmap of the SNPs on the array (to identify possible sample replications or appraise relatedness), a heatmap of the posterior probabilities that the SNP distribution is of the outlier component, scatterplot of first two PCs, boxplots of batch and other characteristics (if indicated) vs the top principle components, csv file of suggested probes to remove based on the percent of samples that failed for that probe (see function details), csv file of suggested samples to potentially remove based on evidence of sex mix-ups, a high proportion of failed probes for that sample, indication of sample contamination, low global methylated or unmethylated intensities, or unintentional replicates based on clustering of SNP probes (see dendogram and heatmap of the SNPs on the array). Also returns a list that includes:

SamplestoRemove	A character vector of Basenames to exclude based on evidence of sex mix-ups, or a high proportion of failed probes for that sample. This vector does not the suspected unintentional replicates; we suggest users check the SNP dendogram to evaluate this themselves.

ProbestoRemove A character vector of CpGIDs to exclude based on the exploratory data analysis

DetectionPval A matrix of detection p-values

IndicatorGoodIntensity
A matrix used to mask methylation values with a poor intensity value based on the number of beads (if minNbeads is larger than zero) and intensity values of zero (if FilterZeroIntensities=TRUE); NA if poor intensity value and 1 otherwise

logOddsContamin
The average log odds from the SNP posterior probabilities from the outlier component; capturing how irregular the SNP beta-values deviate from the ideal trimodal distribution. Values greater than -4 are suggest potentially contaminated samples

Examples

```
## Not run:
EDAtrying<-ExploratoryDataAnalysis(RGset=exampledat,
                                   globalvarexplore=c("Sex","BWT"),
                                   DetectionPvalMethod="SeSAMe",
                                   DetectionPvalCutoff=0.05,
                                   minNbeads=3,
                                   FilterZeroIntensities=FALSE,
                                   destinationfolder="H:/UCLA/PACE/Birthweight-placenta",
                                   savelog=TRUE,
                                   cohort="HEBC",analysisdate="20200710")

## End(Not run)
```

---

loadingSamples *Load Data and Rename Phenotype Data to Standardize*

Description

Reorganize data to standardize for analysis

Usage

```
loadingSamples(
  SamplePlacement = NULL,
  PhenoData = PhenoData,
  IDlink = "ID",
  BWTvar = NULL,
  BATCHvar = NULL,
  SEXvar = NULL,
  FemaleInd = "1",
  MaleInd = "2",
  ETHNICvar = NULL,
  GESTvar = NULL,
  BIRTHLENGTHvar = NULL,
  HEADCIRCUMvar = NULL,
  IDATdir = NULL,
  destinationfolder = NULL,
```

```
  savelog = TRUE,
  cohort = NULL,
  analysisdate = NULL
)
```

Arguments

SamplePlacement	An data.frame that includes 'Basename' and sample ID. Not required if 'Basename'is in PhenoData; might also include an indicator for batch.
PhenoData	A required data.frame that includes the participant characteristics and a sample ID
IDlink	A required character string indicating the column name for the sample ID in the data.frame PhenoData
BWTvar	An optional character string indicating the column name for the birthweight in the data.frame PhenoData; if there is no birthweight variable in data.frame PhenoData, can specify argument value is NULL. Birthweight values should be in grams
BATCHvar	An optional character string indicating the column name for the batch variable in the data.frame PhenoData; if there is no batch variable in the data.frame PhenoData, can specify argument value is NULL. Batch variable is expected to be categorical
SEXvar	A required character string indicating the column name for the infant in the data.frame PhenoData; assumed only two sexes are in the dataset
FemaleInd	A character string indicating how female sex is coded in the dataset; e.g. "1"
MaleInd	A character string indicating how male sex is coded in the dataset; e.g. "2"
ETHNICvar	An optional character string indicating the column name for the maternal race/ethnicity variable in the data.frame PhenoData; if there is no maternal race/ethnicity variable in the data.frame PhenoData, can specify argument value is NULL. Maternal race/ethnicity variable is expected to be categorical
GESTvar	An optional character string indicating the column name for the gestational age variable in the data.frame PhenoData; if there is no gestational age variable in the data.frame PhenoData, can specify argument value is NULL. Gestational age variable is expected to be in weeks
BIRTHLENGTHvar	An optional character string indicating the column name for the birth length variable in the data.frame PhenoData; if there is no birth length variable in the data.frame PhenoData, can specify argument value is NULL. Birth length variable is expected to be in centimeters
HEADCIRCUMvar	An optional character string indicating the column name for the head circumference variable in the data.frame PhenoData; if there is no head circumference variable in the data.frame PhenoData, can specify argument value is NULL. Head circumference variable is expected to be in centimeters
IDATdir	A character string indicating the location of the IDAT files, e.g. "C:/Home/PACE/BirthSize/IDATs"
destinationfolder	A character string indicating the location where files should be saved, e.g. "C:/Home/PACE/BirthSize'
savelog	logic; TRUE indicates to save a log of the functions run (default)
cohort	A character string for the cohort's acronym (e.g. "HEBC")
analysisdate	A character string indicating the date the analysis was run. Please specify in the form: YEARMONTHDAY, e.g. "20200205" for February 5th 2020

Details

This function Renames the variables, puts the variables in the right mode for the analysis, and loads the raw array data.

Value

A RGset (of class RGChannelSetExtended) with raw microarray intensity values. In addition to the columns included in the input PhenoData data.frame, the pData for this RGset will include the following columns:

ID	Sample ID
BWT	Birthweight variable if BWTvar argument was not NULL; numeric
Age	Maternal age variable if AGEvar argument was not NULL; numeric
BMI	Maternal pre-pregnancy variable if BMIvar argument was not NULL; numeric
Sex	Infant sex variable if SEXvar argument was not NULL; factor, 'Female' or 'Male'
Ethnic	Maternal race/ethnicity variable if ETHNICvar argument was not NULL; factor
Gestage	Gestational age variable if GESTvar argument was not NULL; numeric
Batch	Batch variable if BATCHvar argument was not NULL; factor
BirthLength	Birth length variable if BIRTHLENGTHvar argument was not NULL; numeric
HeadCircum	Head circumference variable if HEADCIRCUMvar argument was not NULL; numeric
BWTkg	Birthweight in kg if BWTvar argument was not NULL
wlr	Birthweight birth length ratio if BWTvar and BirthLength arguments were not NULL
BWT_Zscore	Birthweight gestational age and sex-specific Z-scores based on Newborn Cross-Sectional Study of the INTERGROWTH-21st Project
BirthLength_Zscore	Birth length gestational age and sex-specific Z-scores based on Newborn Cross-Sectional Study of the INTERGROWTH-21st Project
HeadCircum_Zscore	Head circumference gestational age and sex-specific Z-scores based on Newborn Cross-Sectional Study of the INTERGROWTH-21st Project
WLR_Zscore	Birthweight birth length ratio gestational age and sex-specific Z-scores based on Newborn Cross-Sectional Study of the INTERGROWTH-21st Project

Examples

```
## Not run:
exampledat<-loadingSamples(SamplePlacement=NULL,PhenoData=allphenodata,IDlink="ID",
                           BWTvar="BWT",BATCHvar="Study",
                           SEXvar="GENDER_A",FemaleInd="1",MaleInd="2",
                           ETHNICvar="ETHNIC",GESTvar="gestAge",
                           BIRTHLENGTHvar="BirthLength",HEADCIRCUMvar=NULL,
                           IDATdir="H:/UCLA/PACE/Birthweight-placenta/IDATfiles",
                           destinationfolder="H:/UCLA/PACE/Birthweight-placenta",
                           savelog=TRUE,
                           cohort="HEBC",analysisdate="20200710")

## End(Not run)
```

`outlierprocess` *Correcting Outliers*

Description

Correcting outliers from the beta-value matrix based on specified approach

Usage

```
outlierprocess(
  processedBetas = NULL,
  quantilemethod = NULL,
  trimming = FALSE,
  pct = 0.005,
  cohort = NULL,
  analysisdate = NULL,
  destinationfolder = NULL
)
```

Arguments

`processedBetas`	The preprocessed beta-value matrix
`quantilemethod`	A character string indicating the approach to estimate quantiles. Two options: 'EmpiricalBeta' or 'Quantile'. To estimate the quantiles based on the empirical beta distribution, specify 'EmpiricalBeta'. To estimate the quantiles using the quantiles function (type7), specify 'Quantile'.
`trimming`	Logic statement (TRUE/FALSE; default is FALSE). If FALSE, winsorize the outliers based on the specified percentile. If TRUE, trims the outliers based on gaps larger than 3*IQR; the cutoff for the number of outliers in a group is either a maximum of 5 or 0.0025 of the total number of samples (whichever is larger). Detected outliers are recoded as NA.
`pct`	Numeric value indicating the percentile winsorized on each side of the distribution; default is 0.005, indicating winsorizing 1 percent.
`cohort`	A character string for the cohort's acronym (e.g. "HEBC")
`analysisdate`	A character string indicating the date the analysis was run. Please specify in the form: YEARMONTHDAY, e.g. "20200205" for February 5th 2020
`destinationfolder`	A character string indicating the location where files should be saved, e.g. "C:\Home\PACE\BirthSize'

Details

If estimating the quantiles based on the empirical beta distribution, the shape parameters are estimated using the method of moments approach. If trimming is TRUE, remove extreme outliers based on gaps that must be at least 3*IQR; the cutoff for the number of outliers in a group is either a maximum of 5 or 0.0025 of the total number of samples (whichever is larger). Detected outliers are recoded as NA.

Value

Returns and saves an RData file of the preprocessed beta-value matrix with the outliers removed or corrected. Also saves a csv of the CpG distributions after removing or correcting the outliers, a csv of the number of outliers per CpG locus, and a csv of the proportion of outliers per sample.

Examples

```
## Not run:
Betasnooutliers<-outlierprocess(processedBetas=betasabovedetection,
                                quantilemethod="EmpiricalBeta",
                                trimming=FALSE,
                                pct=0.005,
                                destinationfolder="H:/UCLA/PACE/Birthweight-placenta",
                                cohort="HEBC",analysisdate="20200710")

## End(Not run)
```

`preprocessingofData` *Perform Sample Pre-processing*

Description

Sample pre-processing so that the data is ready for analysis

Usage

```
preprocessingofData(
  RGset = NULL,
  SamplestoRemove = NULL,
  ProbestoRemove = NULL,
  compositeCellType = NULL,
  KchooseManual = NULL,
  cohort = NULL,
  analysisdate = NULL,
  destinationfolder = NULL,
  savelog = TRUE
)
```

Arguments

`RGset`	The RGset generated by the function loadingSamples. If you want to adjust for batch effects using ComBat, the pData of the RGset must include the column 'Batch'
`SamplestoRemove`	An optional character vector of Basenames to exclude based on the exploratory data analysis; default is NULL
`ProbestoRemove`	An optional character vector of CpGIDs to exclude based on the exploratory data analysis; default is NULL

`compositeCellType`
: A required character string indicating the type of cell composition estimation to perform. The default is "RefFree" indicating a reference-free deconvolution; other options are "Blood", "Placenta", and "CordBlood" to be used with the Houseman reference-based approach. The placenta reference-based approach is based on the data curated in the following publication: https://www.researchsquare.com/article/rs-38223/v3

`KchooseManual`
: Optional numeric value (e.g 2) used to manually set the K (number of cell types) estimated using RefFreeCellMixArray. May need to do this if it seems that there is an extreme outlier driving the number of cell types estimated. See details for more information; the default value is NULL

`cohort`
: A character string for the cohort's acronym (e.g. "HEBC")

`analysisdate`
: A character string indicating the date the analysis was run. Please specify in the form: YEARMONTHDAY, e.g. "20200205" for February 5th 2020

`destinationfolder`
: A character string indicating the location where files should be saved, e.g. "C:\Home\PACE\BirthSize'

`savelog`
: logic; TRUE indicates to save a log of the functions run (default)

Value

Figures (PDFs/PNGs) of associations between top PCs and indicators of batch before and after ComBat (if an indicator for batch was included in the dataset), as well as plots to identify potential outliers driving the estimated cell composition estimates. Also saves an RData file and returns a list that includes:

`mset`
: A GenomicRatioSet subset to the samples and CpG loci among the matrix of processed beta-values

`processedBetas`
: Matrix of the processed beta-values

`Kchoose`
: A numeric value indicating the number of cell types selected by the RefFree method

`Omega`
: A matrix of cell composition estimates generated by the RefFree method

Examples

```
## Not run:
processedOut<-preprocessingofData(RGset=exampledat,
                                  SamplestoRemove=EDAtrying$SamplestoRemove,
                                  ProbestoRemove=EDAtrying$ProbestoRemove,
                                  compositeCellType ="Placenta",
                                  KchooseManual=NULL,
                                  destinationfolder="H:/UCLA/PACE/Birthweight-placenta",
                                  savelog=TRUE,
                                  cohort="HEBC",analysisdate="20200710")

## End(Not run)
```

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